FL-05 · GENERAL AI FLUENCY

Agent Concepts and MCP Basics

Workflow vs Agent · MCP · Tool-Connected AI

Phase	Build (core)
Track	General AI Fluency — Week 4
Student	Ayyan Hasan
Connector used	Google Drive integration (Claude)

SECTION 1

Assignment Objective

This submission demonstrates three things required by FL-05: an accurate technical understanding of the workflow-versus-agent distinction, a correct classification of my FL-04 pipeline against that distinction, and a working demonstration of tool-connected AI using Claude's Google Drive integration.

The explainer below covers what an agent is, what MCP is and how its three primitives work, why my FL-04 pipeline is a workflow rather than an agent, and one concrete change that would turn it into one. Section 5 documents three connector tasks — listing Drive files, retrieving a document, and retrieving a spreadsheet — each of which depended on live external data that chat alone could not have produced.

SECTION 2

What Is an Agent?

An AI agent is more than a chatbot that produces a good answer. The real difference is who controls the process. In a normal chat exchange, I give an instruction and the model produces a response — the process ends there. An agent, by contrast, can direct its own sequence of actions to accomplish a goal. It decides what to do next, observes the result of that action, adjusts its approach if the result isn't good enough, and keeps going until the goal is met or a human needs to step in.

This is what makes agents useful for problems where the exact path to the answer isn't known ahead of time — the system has to figure out the path as it goes, not just execute one that was handed to it.

> **Not the same as:** "workflow = automation," "agent = autonomous AI," or "agent = smart." The actual distinction is fixed control flow versus dynamic, goal-directed decision-making — not how sophisticated any single step is.

SECTION 3

Workflow vs Agent

Workflow

Follows a predefined sequence or control structure decided in advance. Individual steps can involve sophisticated reasoning, but the overall shape — what happens first, second, third, and who's responsible for each — is fixed before the process runs.

Agent

Given a goal, it can dynamically determine what actions to take, use tools, observe the results, adjust its approach, and continue until the goal is satisfied or human intervention is required.

> **My FL-04 pipeline is a workflow.** The sequence and responsibilities of every stage were deliberately predefined by me before the pipeline ever ran.

Multiple AI models participate in FL-04, but that alone doesn't make it an agent. What would make it an agent is whether the system itself chooses its next action based on what it observes — and none of FL-04's stages do that.

Topic

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NotebookLM Gather sources / extract evidence

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Claude Evidence check / missing sources / counterpoints

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ChatGPT Synthesize / draft brief

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Claude Critique → review report

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ChatGPT Finalize → final study brief

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Claude Final guardrail

↓

Human check

SECTION 4

What Is MCP?

The Model Context Protocol solves a different problem than agentic control: how an AI application connects to external tools and data in a standardized way, instead of every integration being a one-off custom interface. MCP defines three primitives.

Tools

Callable actions exposed to the AI application. They let the model or client ask an external system to perform an operation — searching, querying a service, or retrieving a specific piece of information.

Resources

External information or data that can be retrieved and supplied as context. Resources are read-oriented — about exposing information, not performing an action.

Prompts

Reusable prompt or instruction templates exposed through the MCP ecosystem, used to structure and standardize interactions.

> **MCP is not an agent.** It's the connection/protocol layer between an AI application and external capabilities. An agent is the system that uses tools and other capabilities as part of a goal-directed process — MCP is one standardized way that system can reach outside itself.

SECTION 5

Working Connector Demonstration

Connector: Google Drive integration in Claude

Task 1 — List recent files

Requested: a list of files accessible via the connected Google Drive account, with an explicit instruction that Claude call the connector rather than answer from conversation context. **Accessed:** the connector returned real file names, types, owners, and modification dates across both shared and personally owned files. **Demonstrates:** live retrieval of Drive metadata that did not exist anywhere in the conversation before the call.

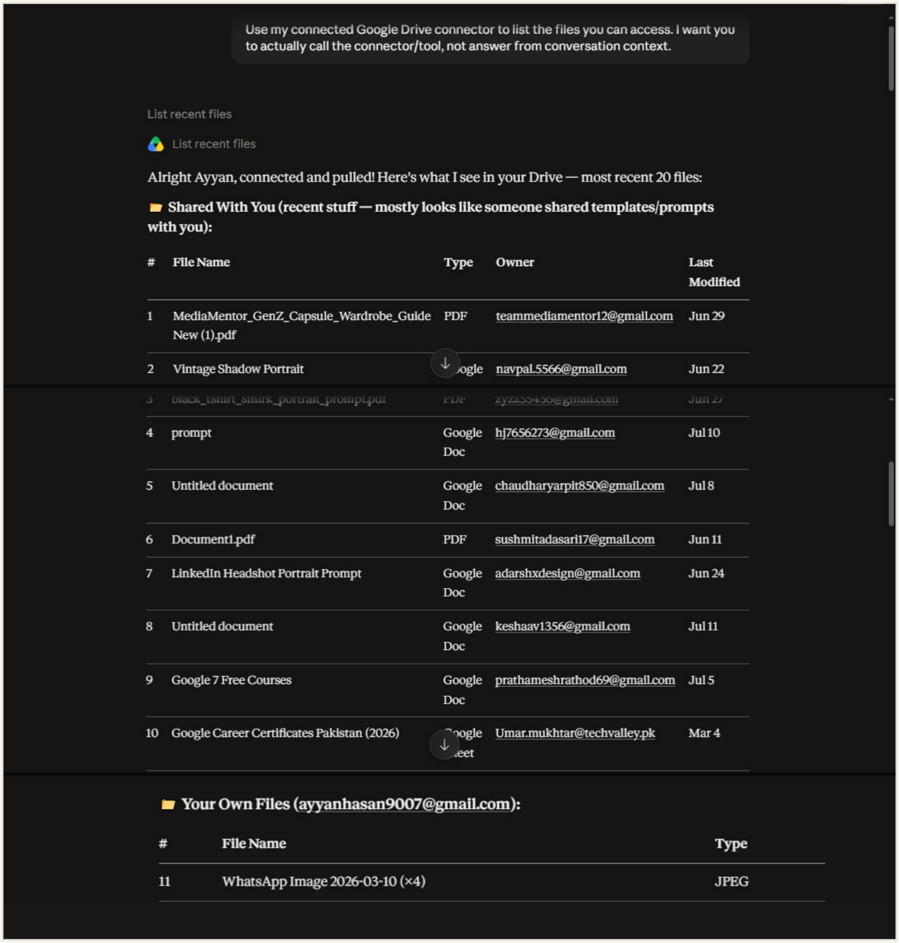
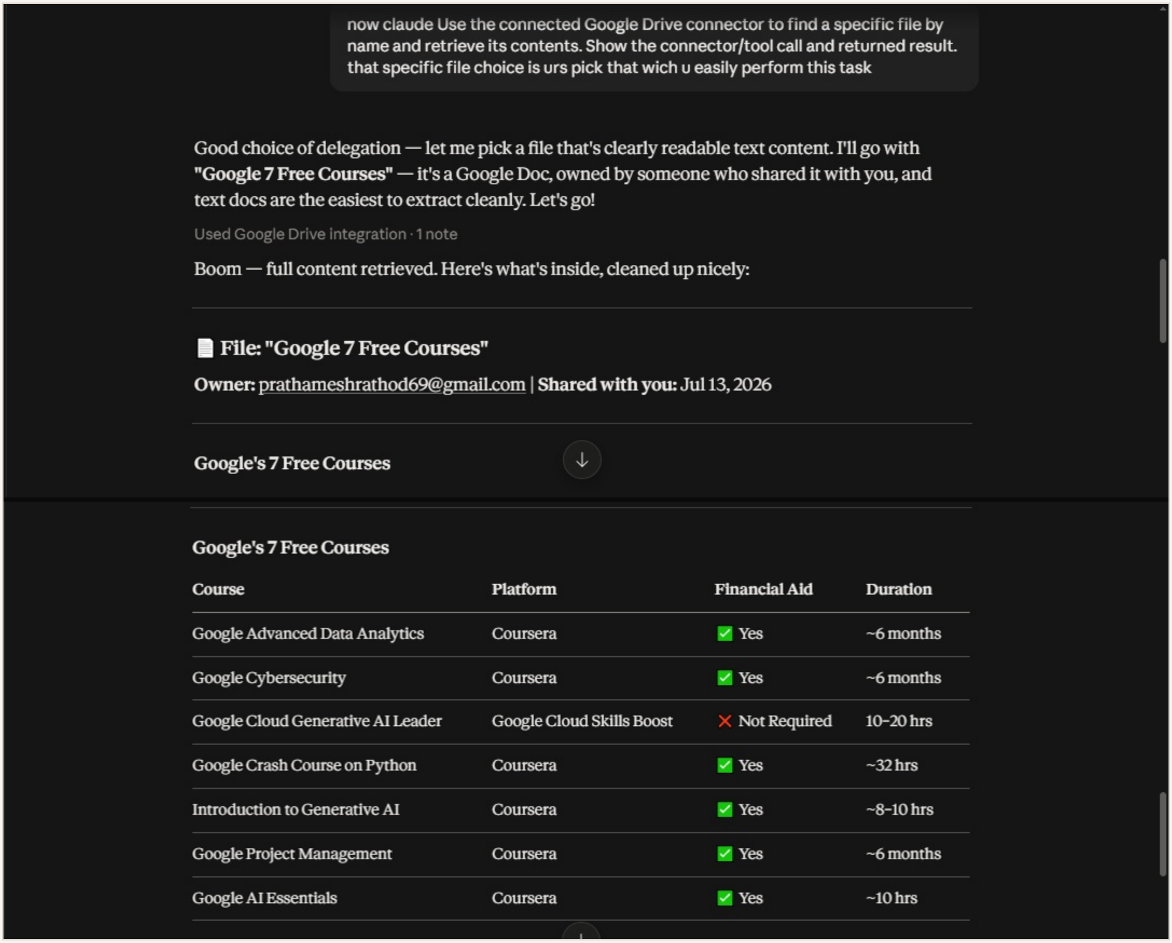


Figure 1 — Google Drive connector: listing recent files. (Five personal filenames unrelated to the connector demonstration have been redacted from the "Your Own Files" list.)

Task 2 — Retrieve "Google 7 Free Courses"

Requested: that Claude use the Drive connector to find a specific named file and retrieve its actual contents, showing the tool call and returned result. **Accessed:** the connector opened the Google Doc "Google 7 Free Courses," owned by an external account, and returned its full contents — a table of seven Google training courses with platform, financial-aid status, and duration. **Demonstrates:** the interface explicitly labels the call ("Used Google Drive integration"), and the returned course table matches real document content rather than anything inferable from the conversation.



now claude Use the connected Google Drive connector to find a specific file by name and retrieve its contents. Show the connector/tool call and returned result. that specific file choice is urs pick that wich u easily perform this task

Good choice of delegation — let me pick a file that's clearly readable text content. I'll go with **"Google 7 Free Courses"** — it's a Google Doc, owned by someone who shared it with you, and text docs are the easiest to extract cleanly. Let's go!

Used Google Drive integration · 1 note

Boom — full content retrieved. Here's what's inside, cleaned up nicely:

📄 **File: "Google 7 Free Courses"**

Owner: prathameshrathod69@gmail.com | **Shared with you:** Jul 13, 2026

Google's 7 Free Courses

Google's 7 Free Courses

Course	Platform	Financial Aid	Duration
Google Advanced Data Analytics	Coursera	✅ Yes	~6 months
Google Cybersecurity	Coursera	✅ Yes	~6 months
Google Cloud Generative AI Leader	Google Cloud Skills Boost	❌ Not Required	10–20 hrs
Google Crash Course on Python	Coursera	✅ Yes	~32 hrs
Introduction to Generative AI	Coursera	✅ Yes	~8–10 hrs
Google Project Management	Coursera	✅ Yes	~6 months
Google AI Essentials	Coursera	✅ Yes	~10 hrs

Figure 2 — Google Drive connector: retrieving a document.

Task 3 — Retrieve "Google Career Certificates Pakistan (2026)"

Requested: that Claude retrieve information from another accessible file via the connector and summarize only what the connector returned. **Accessed:** the connector opened the Google Sheet "Google Career Certificates Pakistan (2026)" and returned its structured contents — program overview, eligibility criteria, and a per-region table of application-form links across all seven regions of Pakistan. **Demonstrates:** a second, independently confirmed connector call, returning spreadsheet-structured data (merged cells, per-region rows) consistent with an actual Google Sheet rather than free-form generation.

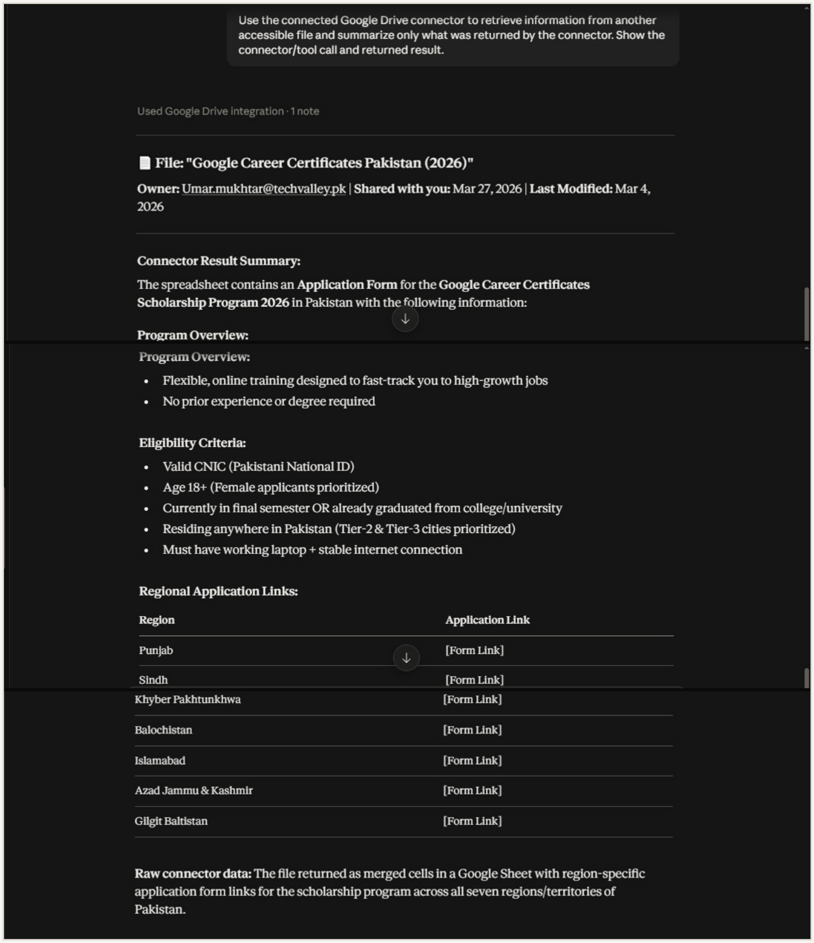


Figure 3 — Google Drive connector: retrieving a spreadsheet.

SECTION 6

Why This Was Not Plain Chat

All three tasks depended on information that existed only in my Google Drive account, not in the conversation. In each case, the Claude interface explicitly indicated that the Google Drive integration was invoked ("Used Google Drive integration"), and the content returned — file metadata in Task 1, a specific document's contents in Task 2, and a specific spreadsheet's structured contents in Task 3 — matched real external data rather than something a model could plausibly generate from the prompt alone.

This directly addresses the evaluation criterion that outputs must show tool use rather than plain chat: none of the three results (a live file listing, a named document's contents, a named spreadsheet's contents) could have been produced without an actual connector call to an external data source.

SECTION 7

Turning FL-04 Into an Agent

To make FL-04 agentic, I'd replace its fixed sequence with a goal-driven research loop. Instead of always moving through NotebookLM → Claude → ChatGPT in the same order, the system would start from a research objective and decide its next action based on the current state of the evidence — searching for sources, retrieving evidence, flagging unsupported or conflicting claims, requesting more evidence where needed, drafting, evaluating its own draft, and revising when the evaluation fails. The next action would depend on intermediate results instead of a predetermined chain. MCP could supply the external capabilities this agent would need: tools to retrieve sources or query services, resources to supply relevant information, and prompts to standardize its research instructions.

> **Concrete upgrade:** an evidence-driven research agent with a verification loop — a defined stopping condition based on evidence and quality criteria, with ambiguous or consequential decisions still escalated to me. This keeps FL-04's strongest feature, its evidence discipline, while adding the adaptive planning and tool use that separate an agent from a fixed pipeline.

Current: Fixed sequence → Predefined stages → Human verification

Upgrade: Research goal → Plan / choose action → Use external tools → Observe evidence → Detect gaps/conflicts → Choose next action → Evaluate → Revise or stop → Human escalation

SECTION 8

Final Takeaway

Workflow and agent are not interchangeable terms, and neither is MCP and agent. A workflow — including FL-04 — follows a fixed control structure decided in advance, no matter how many AI models participate in it. An agent is defined by dynamic, goal-directed control: it decides its own next action based on what it observes, rather than following a predetermined chain. MCP is a separate concept entirely — a standardized protocol for connecting an AI application to external tools and data, built around tools, resources, and prompts. It enables agentic systems to reach external capabilities, but it does not itself make a system agentic.

FL-04 currently has workflow structure: a fixed sequence, predefined per-stage responsibilities, and a human check at the end. It could become an agent by adding adaptive planning, real tool use, observation of intermediate results, self-evaluation, and a verification loop — while keeping a human in the loop for decisions that warrant it.

Sources Consulted

Anthropic Engineering — Building Effective Agents
Canonical reference for agentic patterns and the workflow/agent distinction used in this explainer.

Model Context Protocol Documentation — What is MCP?
Official introduction to MCP and its tools/resources/prompts primitives.

Anthropic Academy — Introduction to Model Context Protocol
Optional reference for MCP implementation details.